Reflect with us - Week 5

This assignment will not be graded and is only for practice.

1 point

Consider a map $T: R^2 \to R^2$ defined as $T(v) = Av$, where $v = \begin{bmatrix} x \\ y \end{bmatrix}$, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $det(A) \neq 0$. Which of the following options are correct?

- [ ] T is a linear transformation.
- [ ] T is both one-one and onto.
- [ ] T is neither one-one nor onto.
- [ ] T is one-one but not onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is a linear transformation.

T is both one-one and onto.

- Step 1:

1 point

What is $T(v_1 + v_2)$?

○ $A(v_1 + v_2)$.

○ Av_1v_2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$A(v_1 + v_2)$.

Recall: $A(v_1 + v_2) = Av_1 + Av_2$

- Step 2:

1 point

Is $T(v_1 + v_2) = T(v_1) + T(v_2)$?

○ Yes.

○ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Check: $T(cv) = cT(v)$.

- Hence T is linear transformation. So Option 1 is true.
- Another method: Write the definition of T explicitly as,

$$T(x, y) = (ax + by, cx + dy)$$

- Step 1:Step 1: What is $T((x_1, y_1) + (x_2 + y_2))$T((x1, y1) + (x2 + y2))?

$$\begin{aligned} T(x_1 + x_2, y_1 + y_2) &= (a(x_1 + x_2) + b(y_1 + y_2), c(x_1 + x_2) + d(y_1 + y_2)) \\ &= (ax_1 + ax_2 + by_1 + by_2, cx_1 + cx_2 + dy_1 + dy_2) \\ &= (ax_1 + by_1, cx_1 + dy_1) + (ax_2 + by_2, cx_2 + dy_2) \\ &= T(x_1, y_1) + T(x_2, y_2) \end{aligned}$$

T(x1 + x2, y1 + y2) = (a(x1 + x2) + b(y1 + y2), c(x1 + x2) + d(y1 + y2)) = (ax1 + ax2 + by1 + by2, cx1 + cx2 + dy1 + dy2) = (ax1 + by1, cx1 + dy1) + (ax2 + by2, cx2 + dy2) = T(x1, y1) + T(x2, y2)

- Step 2:Step 2:

  1 point

  Is $T(c(x, y)) = cT(x, y)$T(c(x, y)) = cT(x, y)?

  ◯ Yes.

  ◯ No.

  No, the answer is incorrect.

  Score: 0

  Accepted Answers:

  Yes.

- Hence TT is linear transformation. So Option 1 is true.

Now in the given problem $T(v) = 0$T(v) = 0, implies $Av = 0$Av = 0. This is the matrix representation of the system of linear equations:

$$\begin{aligned} ax + by &= 0 \\ cx + dy &= 0 \end{aligned}$$

ax + bycx + dy = 0 = 0

- Step 3:Step 3:

  1 point

  If the system of linear equations $Av = 0$Av = 0 has a unique solution, then what is the solution?

  ◯

  $v = 0$v = 0

  ◯

  vv can be any vector in R^2 R2.

  No, the answer is incorrect.

  Score: 0

  Accepted Answers:

  $v = 0$v = 0

- Step 4:Step 4:

  1 point

  What is the condition on AA, for which the system of linear equations $Av = 0$Av = 0 has a unique solution? [Recall from Weeks 1 and 2]

  ◯

  $det(A) = 0$det(A) = 0.

  ◯

  $det(A) \neq 0$det(A) = 0.

  No, the answer is incorrect.

  Score: 0

  Accepted Answers:

  $det(A) \neq 0$det(A) = 0.

Concluding, we can say that, if $det(A)$det(A) is non-zero, then $T(v) = 0$T(v) = 0 implies $v = 0$ v = 0. Hence TT is one to one.

- Let $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \in R^2$. We are going to find whether there exists a vector $v = \begin{bmatrix} x \\ y \end{bmatrix} \in R^2$, such that $T(v) = Av = w$.

  $Av = w$ is the matrix representation of the system of linear equation:

  $$\begin{matrix} ax + by = w_1 \\ cx + dy = w_2 \end{matrix}$$

- Step 5:Step 5:

  1 point

  If $det(A) \neq 0$, then what can we say about the solution of the above system of linear equations?

  ○ No solution.

  ○ Unique solution.

  No, the answer is incorrect.

  Score: 0

  Accepted Answers:

  Unique solution.

  Concluding from this, we can say that, if $det(A) \neq 0$, then T is onto.

  Hence, Options 1 and 2 are the correct options.Hence, Options 1 and 2 are the correct options.

  Check Answers

  Your score is: 0/7